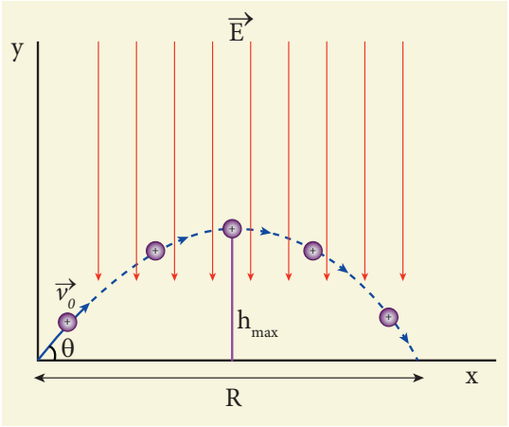
EXAMPLE 1.19

A small ball of conducting material having a charge +q and mass m is thrown upward at an angle θ to horizontal surface with an initial speed v_o as shown in the figure. There exists an uniform electric field E downward along with the gravitational field g. Calculate the range, maximum height and time of flight in the motion of this charged ball. Neglect the effect of air and treat the ball as a point mass.



Solution

If the conductor has no net charge, then its motion is the same as usual projectile motion of a mass m which we studied in Kinematics (unit 2, vol-1 XI physics). Here, in this problem, in addition to downward gravitational force, the charge also will experience a downward uniform electrostatic force.

The acceleration of the charged ball due to gravity = $-g\,\hat{j}$

The acceleration of the charged ball due to uniform electric field = $-\frac{qE}{m}\hat{j}$

The total acceleration of charged ball in downward direction $\vec{a} = -\left(g + \frac{qE}{m}\right)\hat{j}$

It is important here to note that the acceleration depends on the mass of the object. Galileo's conclusion that all objects fall at the same rate towards the Earth is true only in a uniform gravitational field. When a uniform electric field is included, the acceleration of a charged object depends on both mass and charge.

But still the acceleration $a = \left(g + \frac{qE}{m}\right)$ is constant throughout the motion. Hence we use kinematic equations to calculate the range, maximum height and time of flight. In fact we can simply replace g by $g + \frac{qE}{m}$ in the usual expressions of range, maximum height and time of flight of a projectile.

	Without charge	**With the charge +q**
Time of flight T	$\frac{2v_\circ \sin\theta}{g}$	$\frac{2v_\circ \sin\theta}{\left(g + \frac{qE}{m}\right)}$
Maximum height h_{max}	$\frac{v_\circ^2 \sin^2\theta}{2g}$	$\frac{v_\circ^2 \sin^2\theta}{2\left(g + \frac{qE}{m}\right)}$
Range R	$\frac{v_\circ^2 \sin 2\theta}{g}$	$\frac{v_\circ^2 \sin 2\theta}{\left(g + \frac{qE}{m}\right)}$

Note that the time of flight, maximum height, range are all inversely proportional to the acceleration of the object. Since $\left(g + \frac{qE}{m}\right) > g$ for charge +q, the quantities T, h_{max}, and R will decrease when compared to the motion of an object of mass m and zero net charge. Suppose the charge is –q, then $\left(g - \frac{qE}{m}\right) < g$, and the quantities T, h_{max} and

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